

Supplementary Material: Exact Radial-Determinant Flexural Frequencies of a Monolithic Thickness-Poled Piezoelectric Ring: An Open-Circuit-Equals-Short-Circuit Theorem and a Segmented-Electrode Sensing Admittance

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Main text contains no job numbers; they are listed in §S.8.

S.1 Cubic consistency condition and the open-circuit-equals-short-circuit theorem

S.1.1 Short-circuit potential and the projected electrical equation

With both faces grounded, $\phi(\pm H) = 0$, write $\phi = \bar{\phi}(r, \theta, t)f(z)$ with $f(\pm H) = 0$, so $E_z = -\bar{\phi}f'$ and $E_{\parallel} = -f\nabla\bar{\phi}$. With $\varepsilon_{rr} + \varepsilon_{\theta\theta} = -z\nabla^2 w$, the electric enthalpy $\mathcal{H} = \frac{1}{2}\varepsilon \cdot \bar{c}\varepsilon - \bar{e}_{31}E_z(\varepsilon_{rr} + \varepsilon_{\theta\theta}) - \frac{1}{2}\bar{\Xi}_{33}E_z^2 - \frac{1}{2}\bar{\Xi}_{11}|E_{\parallel}|^2$ integrates through the thickness to

$$\mathcal{H}_2 = U_{\text{el}}(w) - \bar{e}_{31}I_1 \bar{\phi} \nabla^2 w - \frac{1}{2}\bar{\Xi}_{33}I_2 \bar{\phi}^2 - \frac{1}{2}\bar{\Xi}_{11}I_3 |\nabla\bar{\phi}|^2, \quad (\text{S.1})$$

with $I_1 = \int z f' dz$, $I_2 = \int f'^2 dz$, $I_3 = \int f^2 dz$ over $|z| \leq H$. Stationarity in w and in $\bar{\phi}$ (Hamilton's principle, the same weight for both fields) gives the flexural equation (A) below with $K_{\text{pref}} = -\bar{e}_{31}I_1$, the electrical equation $\bar{\Xi}_{11}I_3 \Delta\bar{\phi} - \bar{\Xi}_{33}I_2 \bar{\phi} + K_{\text{pref}}\Delta w = 0$, and the natural rim condition $\partial\bar{\phi}/\partial r = 0$. In the long-wave limit $\bar{\phi} \rightarrow (K_{\text{pref}}/\bar{\Xi}_{33}I_2)\Delta w$ and $D_{\text{SC}} - d = (\bar{e}_{31}I_1)^2/(\bar{\Xi}_{33}I_2)$. By Cauchy–Schwarz, $I_1^2 \leq I_2 \int z^2 dz = \frac{2}{3}H^3 I_2$, with equality only for $f' \propto z$: the quadratic $f = 1 - z^2/H^2$ is the unique one-term profile whose long-wave increment is the exact thin-plate value $\frac{2}{3}H^3 \bar{e}_{31}^2/\bar{\Xi}_{33}$ (from $\partial_z D_z = 0$ with $\phi(\pm H) = 0$). Its integrals are $I_1 = -4H/3$, $I_2 = 8/(3H)$, $I_3 = 16H/15$, so

$$M_{rr}^{\text{piezo}} = -\bar{e}_{31} \int_{-H}^H z E_z dz = -\frac{4H}{3} \bar{e}_{31} \bar{\phi}, \quad K_{\text{pref}} = \frac{4}{3} H \bar{e}_{31}. \quad (\text{S.2})$$

For comparison, the sinusoidal profile $f = \cos(\pi z/2H)$ gives $K_{\text{pref}} = (4H/\pi)\bar{e}_{31}$ and a long-wave increment $96/\pi^4$ of exact under this projection; closing its electrical equation instead by the thickness-integrated Gauss law (Duan, Quek and Wang's Eq. (12) [1]) replaces $\bar{\Xi}_{33}I_2\bar{\phi}$ by the face jump of D_z and gives an increment $12/\pi^2 = 1.216$ times exact. That closure was used in an earlier version

of this model; its free-free split (+2.347% against +1.939%) is the 83% finite-element recovery discussed in §S.3. The full moment and shear are

$$M_{rr} = -\left[d\partial_r^2 w + (d - 2A_1)(r^{-1}\partial_r w + r^{-2}\partial_\theta^2 w) + K_{\text{pref}}\bar{\phi}\right], \quad Q_r = -\left[d\partial_r \nabla^2 w + K_{\text{pref}}\partial_r \bar{\phi}\right], \quad (\text{S.3})$$

with twisting moment $M_{r\theta} = -2A_1 \partial_r(r^{-1}\partial_\theta w)$, which carries no piezoelectric term. The free-edge natural conditions are $M_{rr} = 0$ and the Kirchhoff effective shear $Q_r^{\text{eff}} = Q_r + r^{-1}\partial_\theta M_{r\theta} = 0$. On a Helmholtz branch $\Delta Z = \lambda Z$ at angular order n this is $Q_r^{\text{eff}}[cZ] = -[KZ' - 2A_1 n^2(Z'/r^2 - Z/r^3)]$, with $K = d\lambda + K_{\text{pref}}\chi$. It reduces to Q_r at $n = 0$. The flexural motion equation is

$$d\Delta\Delta\bar{w} + K_{\text{pref}}\Delta\bar{\phi} - A_2\omega^2\bar{w} = 0. \quad (\text{A})$$

The projected electrical equation, divided by $\bar{\Xi}_{33}I_2$, is

$$a\Delta\bar{\phi} - \bar{\phi} + c\Delta\bar{w} = 0, \quad a = \frac{2H^2\bar{\Xi}_{11}}{5\bar{\Xi}_{33}}, \quad c = \frac{H^2\bar{e}_{31}}{2\bar{\Xi}_{33}}. \quad (\text{B})$$

Positing $\bar{\phi} = \chi\bar{w}$ with χ independent of r , (B) forces $\Delta\bar{w} = \lambda_B\bar{w}$ with $\lambda_B = \chi/(a\chi + c)$, and (A) forces $d\lambda^2 + K_{\text{pref}}\chi\lambda - A_2\omega^2 = 0$. Eliminating λ between the two gives a cubic in χ :

$$\begin{aligned} a_3\chi^3 + a_2\chi^2 + a_1\chi + a_0 &= 0, \\ a_3 &= K_{\text{pref}}a, \quad a_2 = d + K_{\text{pref}}c - A_2\omega^2a^2, \\ a_1 &= -2A_2\omega^2ac, \quad a_0 = -A_2\omega^2c^2. \end{aligned} \quad (\text{S.4})$$

Three roots $\chi_i(\omega)$ give three Helmholtz branches $\Delta\bar{w} = \lambda_i\bar{w}$, two Bessel functions per branch: six constants, matching the sixth-order coupled system. $\chi = 0$ is a root only at $\bar{e}_{31} = 0$ ($a_0 \propto \bar{e}_{31}^2\omega^2$), which is why the purely elastic ring is assembled as a separate 4×4 (§S.2) rather than by zeroing $\bar{e}_{31}$ inside this cubic. These coefficients are *not* obtained from the layered plate's own cubic [3] by the substitution $h_1 \rightarrow 2H$: the two potential shapes have opposite parity about their respective plate midplanes (this ansatz is even in z about the monolithic plate's own midplane; the layered plate's per-skin interior profile is neither even nor odd about the sandwich midplane), so the two cubics are genuinely different operators sharing only their general architecture.

The insulated-rim electrical condition, $\partial\bar{\phi}/\partial r = 0$ at r_i and r_o , together with two mechanical rows per edge, fills the 6×6 . Free-free uses $M_{rr} = Q_r^{\text{eff}} = 0$ at both edges; clamped-clamped uses $w = w' = 0$; clamped-free and free-clamped use one of each pairing at the two edges (§S.6).

S.1.2 Open-circuit and the coincidence theorem

With the bus ansatz $\phi = Vz/(2H) + \bar{\phi}(1 - z^2/H^2)$, the linear term's field $E_z^{\text{lin}} = -V/(2H)$ is spatially constant – even in z – so $\int_{-H}^H z\sigma^{\text{lin}}dz = 0$ identically: the linear piece never enters M_{rr} , for any V , H , or material constant. Because the bus is a single equipotential surface with no r, θ dependence, $E_{\parallel}^{\text{lin}} = 0$, and the enthalpy cross term $-\bar{\Xi}_{33}E_z^{\text{lin}}E_z^f$ integrates with weight $\int_{-H}^H f' dz = f(H) - f(-H) = 0$; so the linear piece drops from both (A) and (B): the open-circuit eigenvalue problem for $(w, \bar{\phi})$ is exactly the short-circuit pair, and V is fixed only afterward, by the scalar charge constraint $Q = 0$. The bus charge is the variable conjugate to V in the same enthalpy,

$$Q = -\frac{\partial}{\partial V} \int_A \mathcal{H}_2 dA = -C_0V, \quad C_0 = \frac{\bar{\Xi}_{33}A}{2H}, \quad (\text{S.5})$$

with no motional term: the strain coupling of the linear field carries the weight $\int_{-H}^H z dz = 0$, and $\bar{\phi}$ does not enter by the cross-term argument above. Hence $V = 0$ on an open-circuit bus for every flexural mode, and open-circuit coincides with short-circuit. (Integrating Gauss’s law pointwise to the face instead gives a charge with motional terms in $\int_A \nabla^2 w$ and $\int_A \bar{\phi}$; for the full face these cancel, the full-face $Q \equiv 0$ result of §S.7, so the two routes agree on the full electrode.) This is the theorem stated in the main text: because V never enters M_{rr} – not merely on a special mechanical edge, as on the layered plate, but identically – open-circuit flexure coincides with short-circuit flexure for every mechanical boundary condition, not only clamped–clamped. On the layered plate the coincidence is a corollary of $S = 0$ under clamped–clamped conditions specifically [3]; here it holds even when $S \neq 0$, because the moment channel for V does not exist at all on this stacking.

S.2 Elastic-limit gate

Setting $\bar{e}_{31} = 0$, $\bar{\phi} = 0$ in (A) recovers the isotropic Kirchhoff ring equation with $D = d = \frac{2}{3}\bar{c}_{11}^E H^3$, $\nu = \nu_p = \bar{c}_{12}^E/\bar{c}_{11}^E$, plate thickness $h = 2H$. Matching the standard isotropic bending stiffness $D = Eh^3/[12(1 - \nu^2)]$ requires $E = \bar{c}_{11}^E(1 - \nu_p^2)$. At the material point of the main text this gives $\bar{c}_{11}^E = 85.661$ GPa, $\bar{c}_{12}^E = 24.661$ GPa, $\nu_p = 0.28789$, $E = 78.561$ GPa. The free–free $n = 0$ fundamental of the new mixed-edge 4×4 at these constants agrees with an independently implemented closed-annulus Kirchhoff solver [2] at 4.77×10^{-10} relative. The coupled 6×6 is not exercised at this gate (§S.1: $\chi = 0$ is a root only at $\bar{e}_{31} = 0$, so the coupled operator is the wrong tool at exact zero coupling). This gate passed before any coupled number in the main text was trusted, and uses no piezoelectric code path.

S.3 Finite-element cross-check

Axisymmetric coupled-field finite elements (piezoelectric-plane elements, KEYOPT(1)=1001, Lanczos eigensolver), millimetre–newton– second–tonne units with relative permittivity, one PZT-4 rectangle $|z| \leq H = 10$ mm (no steel host), cloned from the mesh, material block, and electrical-boundary conventions already confirmed for the layered plate’s own free–free family [3], with the host volume deleted and the piezo volume filling the full thickness. Three decks on the same mesh: coupling-off (e0), both-faces short-circuit (scboth, VOLT= 0 at $z = \pm H$), and open-circuit (oc, VOLT= 0 at neither face, a single coupled VOLT set on both faces together, unconstrained). Job **2520476**, compute-33-02.mill, 3/3 RUN COMPLETED, **0 errors, 0 warnings** on every deck, scored from the raw transcript and mode/discriminator files.

Mode 1 is the free–free rigid body (0 Hz) on every deck. Mode 2 is the first bending mode, identified by the through-thickness discriminator at $r = r_o$: transverse displacement same sign at $z = \pm H$, in-plane displacement opposite sign, ratio ≈ 19.7 on all three decks.

Table S.1: F–F $n = 0$ finite-element mode 2 (bending), job 2520476.

deck	frequency [Hz]	closed form [Hz]	rel. vs closed form
e0 (coupling off)	73.506288	73.593	–0.118%
scboth (SC)	74.933783	75.020	–0.115%
oc (OC)	74.933783	75.020	–0.115%

scboth and oc mode 2 are bit-identical at the printed eight significant figures, 74.933783 Hz on both – the coincidence theorem confirmed in three-dimensional coupled-field finite elements, not only in the Kirchhoff 6×6 – and so are the next several flexural overtones on the same pair of

decks. `scboth` sits 1.942% above `e0`, against a closed-form +1.939% (ratio 1.001); the finite-element short-circuit frequency sits -0.115% from the closed form, the same offset as the coupling-off deck (-0.118%), so the residual is the elastic Kirchhoff-versus-2D difference, not the coupling. **History.** The first version of this model closed the electrical equation by the thickness-integrated Gauss law with a sinusoidal profile (§S.1) and predicted +2.347% here; finite elements recovered 83% of it. The same closure predicted +0.961% for a clamped PZT-4 annular sector ($r_i/r_o = 0.03/0.07$, $2H = 0.8\text{ mm}$, $2\Theta = \pi$), where three-dimensional SOLID226 finite elements give +0.782% (ratio 0.814); the consistent projection gives +0.796% for that sector. Both ratios are $\pi^2/12 = 0.822$ to within the elastic offsets, which is how the projection was identified. Neither comparison was used to set a coefficient: the quadratic and its projection are fixed by the derivation of §S.1.

A companion extensional mode on the same mesh (mode 6 on `e0` and `scboth`) is 1669.57 Hz on both, and 1699.58 Hz on `oc` (+1.80%): the linear (membrane) piezoelectric field does couple to extension, confirming the negative control of main-text §2.1 in finite elements as well as in closed form. This is disclosed as claim C of this series [3] and is not built further here.

S.4 e_{31} inverse: gates, sensitivity, reflection twin

With $\bar{c}_{11}^E$ held at its known value, e_{31} is inverted from the free-free (`coupled_bisect`) and clamped-clamped (`cc_coupled_bisect`) short-circuit fundamentals of Table 1 of the main text. Job **2520482**, `compute-33-02.mill`, 562.0 s, `dps= 60/inner_iters= 45/ outer_iters= 30`, `.err` empty (0 bytes).

G_worker. The worker-process scalar reconstruction reproduces the in-process bisection roots at `rel= 0.0` on both legs, before any inversion built on the in-process class is trusted.

G0 (round trip). Bracket $[1.0, 8.0]\text{ C m}^{-2}$ on the physical branch (see the reflection-twin derivation below for why this range, not $[1, 10]$). Recovered $\hat{e}_{31} = 4.100000001$ against true 4.1, `rel_err =  $2.499 \times 10^{-10}$` (bar 10^{-6}), identical on both legs.

G1 (negative controls). A wrong-sign bracket $[-8.0, -1.0]$ shows no sign flip; $e_{31} = -4.1$ gives a frequency $> 5\%$ from target on both legs; $e_{31} = 0$ raises the singular-cubic exception of §S.1 when routed through a real determinant call (not at construction alone).

G2 (sensitivity). F-F: $d\omega/de_{31} = -3.617563\text{ rad s}^{-1}$ per C m^{-2} ; propagated uncertainty 0.3178% of e_{31} at 0.01% frequency precision, 3.1780% at 0.1%. C-C: $-12.59036\text{ rad s}^{-1}$ per C m^{-2} ; 0.3405% and 3.4050%. Both legs agree closely with the independent long-wave closed-form estimate of main-text §2.4 ($\sim 0.3\%$ of e_{31} at 0.01% frequency if stiffness is known). (The Gauss-law closure of §S.1 gave 0.2649% and 0.2822%; its larger split made e_{31} look better conditioned, and fitted to a measured or finite-element split it would return $|\bar{e}_{31}|$ low by $(\pi^2/12)^{1/2} = 0.907$.)

Reflection twin, derived. Let $\bar{e}_{31} = e_{31} - (c_{13}^E/c_{33}^E)e_{33}$. The cubic coefficients of (S.4) are odd in $\bar{e}_{31}$ for a_3, a_1 and even for a_2, a_0 (K_{pref} and c are odd, a is even). Flipping $\bar{e}_{31} \rightarrow -\bar{e}_{31}$ therefore maps the root set $\{\chi_i\} \rightarrow \{-\chi_i\}$ while each λ_i is invariant (numerator and denominator of $\lambda_B(\chi)$ flip together) and $K_{\text{pref}}\chi$ is invariant termwise (K_{pref} and χ each flip, the product does not): the determinant's M -row and Q -row blocks are unchanged, and only the $\bar{\phi}'$ -row block flips sign as a whole – an overall row rescaling that does not move where the determinant vanishes. Hence $e_{31}^{\text{twin}} = 2(c_{13}^E/c_{33}^E)e_{33} - e_{31}$ gives a bit-identical root frequency to e_{31} itself. At $e_{31} = 4.1$, $e_{31}^{\text{twin}} = 13.800869565217392$, confirmed to `rel= 0` on both legs. This is why the inversion bracket is $[1.0, 8.0]$ and not a wider $[1, 10]$: a wider bracket would cross $\bar{e}_{31} = 0$ ($e_{31} \approx 8.9504\text{ C m}^{-2}$, where the cubic degenerates in degree) and approach the twin itself. Do not chase the twin as a second physical solution; it is the same root, not a different one.

S.5 Overtones and thickness sweep

A sign-flip scan (dps= 60/inner_iters= 45, production settings; job **2521175**, compute-34-07.mill, 421.8 s, .err empty) found three F-F pairs in $[455, 6000] \text{ rad s}^{-1}$ and two C-C pairs in $[1700, 6000]$; every pair reproduces the reflection twin of §S.4 at $\sim 5\text{--}6 \times 10^{-16}$ relative, not only at the fundamental.

Table S.2: F-F and C-C $n = 0$ overtones. Elastic and short-circuit (= open-circuit) roots in rad s^{-1} .

edge	mode	elastic	SC = OC	split
F-F	fundamental	462.398	471.365	+1.939%
F-F	OT1	2169.893	2203.925	+1.568%
F-F	OT2	5270.649	5355.495	+1.610%
C-C	fundamental	1726.890	1757.694	+1.784%
C-C	OT1	4776.965	4861.470	+1.769%

Short-circuit sits above elastic at every pair, both legs – the main-text stiffening-sign theorem generalizes beyond the fundamental. The split trend is not monotonic: F-F dips from 1.94% to 1.57% at the first overtone, then ticks back up to 1.61% at the second; C-C stays flatter, 1.78% $\rightarrow$ 1.77%.

A nine-point thickness sweep (F-F $n = 0$ only, fixed r_i, r_o , job **2521176**, same host, concurrent, 805.8 s, .err empty) tests the long-wave prediction that the split is independent of H at leading order.

Table S.3: F-F $n = 0$ fundamental versus ceramic thickness, fixed radii. Long-wave closed-form estimate of the thin-limit split: 1.814%.

H/r_o	elastic [rad/s]	SC [rad/s]	split
0.005	138.719	141.410	1.9399%
0.010	277.439	282.820	1.9397%
0.01667	462.398	471.365	1.9392%
0.025	693.596	707.040	1.9383%
0.040	1109.754	1131.234	1.9355%
0.060	1664.632	1696.760	1.9300%
0.090	2496.947	2544.843	1.9182%
0.130	3606.702	3675.111	1.8967%
0.180	4993.895	5086.967	1.8637%

Every point returns exactly one elastic and one coupled root in its search window; short-circuit exceeds elastic at every point; the thinnest point's split (1.9399%) is within a factor of 1.069 of the raw long-wave estimate $\frac{1}{2}\bar{c}_{31}^2/(\bar{\Xi}_{33}\bar{c}_{11}^E) = 1.8141\%$ (unrounded constants; the ratio exceeds one because the same $\bar{c}_{31}^2/\bar{\Xi}_{33}$ also stiffens $\bar{c}_{12}^E$, which the bending-only estimate omits). The H -independence prediction holds well below $H/r_o \approx 0.025$ (split flat near 1.94%) and then drifts down to 1.8637% by $H/r_o = 0.18$, a -3.93% relative drift from the thinnest point – a real, small finite-thickness correction beyond the long-wave leading order, consistent with Kirchhoff kinematics being best justified in exactly the thin regime the main-text geometry ($H/r_o \approx 0.0167$) already sits in.

S.6 Mixed-edge (C–F/F–C) bit-identical reduction proof

Clamped–free and free–clamped determinants are built from a single mixed-edge row-selection routine that picks, at each edge independently, either the (w, w') pair (clamped) or the $(M_{rr}, Q_r^{\text{eff}})$ pair (free), following the pattern already used for the elastic ring’s own mixed-edge family [2], adapted to the three-branch cubic of §S.1. Because free–free and clamped–clamped are literal special cases of this routine (both edges set to the same label), the reduction was checked bit-for-bit (`mpmath` value equality, not a tolerance), not only re-derived: at $n = 0$, $\omega = 500$ and $n = 2$, $\omega = 1500$, the mixed-edge routine with both edges ‘F’ (respectively ‘C’) reproduces the original free–free (respectively clamped–clamped) elastic and coupled determinants exactly, eight of eight comparisons. This is a stronger correctness guarantee than an independent derivation would give: a defect in the row-selection logic would almost certainly have broken the already-validated free–free and clamped–clamped paths too, not only left a silent error in the new clamped–free/free–clamped paths.

Physical anchors (job **2521220**, `compute-34-04.mill`, 369.0 s, `.err` empty): clamped–free elastic = 1605.4427637454191, SC = 1637.9221255770667 (+2.023%); free–clamped elastic = 556.8844158380273, SC = 564.9003213461377 (+1.439%). Both stiffen, consistent with the main text’s stiffening-sign estimate (Eq. 7). The negative control ($e_{31} = 0$ refused) still fires through the new coupled methods. A worker-path regression – new scalar-reconstruction functions for each of the four new bisection methods – was confirmed bit-identical to in-process, both in a standalone check and in the local test suite, for all four new methods.

The genuine physical finding of main-text §4.2 – the inner edge dominating bending stiffness far more than the outer edge at this $r_i/r_o = 1/6$ – is reported as an empirical fact of this one geometry, with a plausible but unproven mechanism: at $n = 0$ the free-edge moment condition’s r -dependent term is $-(2A_1/r) Z'$, which is $r_o/r_i = 6\times$ more sensitive to a clamp override at r_i than at r_o , consistent with, but not a rigorous proof of, the observed asymmetry (clamped–free at 90.4% of the way from free–free to clamped–clamped; free–clamped at 7.5%).

S.7 Force-driven sensing admittance: construction and gates

The two-region twelve-unknown system of the main text’s numerical-method section is built from the same three χ -cubic branches as §S.1, split at the load radius r_F , with column order (region I then region II, three branches, two Bessel functions each) matching the free-vibration determinant’s own inner loop. On a Helmholtz function $\Delta Z = \lambda Z$, the physical $n = 0$ resultants collapse to $M_{rr}[cZ] = -(KZ - 2A_1Z'/r)$, $Q_r[cZ] = -KZ'$, $K = d\lambda + K_{\text{pref}}\chi$. Six rim rows (three per edge, F–F SC) plus six interface rows – $[w] = [w'] = [M_{rr}] = [\bar{\phi}] = [\bar{\phi}'] = 0$ and the one inhomogeneous shear jump $[Q_r] = -F/(2\pi r_F)$ – close the twelve-unknown system, solved by an equilibrated linear solve (row and column scaling, tracked and un-scaled after solving) rather than a determinant search, because the third (evanescent) χ -branch produces radial arguments large enough that raw entries span more than fifty orders of magnitude at this geometry.

Five gates, all in-sandbox and independently cluster-confirmed (job **2521701**, `compute-34-04.mill`, 247.5 s, `.err` empty, every printed and logged number bit-identical between sandbox and cluster):

- **G1, force-balance identity** (the load-bearing sign gate on the shear-jump right-hand side): $2\pi \int_{r_i}^{r_o} r w(r) dr = -F/(A_2\omega^2)$ at `rel_err = 1.112761845... × 10−61` – essentially exact, confirming the jump’s sign without assuming it.

- **G2, pole location** (unique-continuation gate): the driven system’s own homogeneous determinant and the free-vibration short-circuit bisection agree at the F–F fundamental, $471.3646021920983 \text{ rad s}^{-1}$, $\text{rel_err} = 0.0$ on every computed digit.
- **G3, $\bar{e}_{31} \rightarrow 0$ limit**, scaled correctly (raw e_{31} and e_{33} together, so $\bar{e}_{31}$ itself scales to zero – see the note below): Q_{in} at scale $1, 10^{-1}, \dots, 10^{-4}$ is 5.004×10^{-12} , 6.627×10^{-13} , 6.648×10^{-14} , 6.648×10^{-15} , 6.648×10^{-16} ; decade ratios 7.551, 9.968, 10.000, 10.000; log–log slope over the three deep decades is 0.999993, clean linear vanishing.
- **G_mu** (diagnostic): the two mechanical Helmholtz branches match the purely elastic wavenumber $\pm\mu = \pm 10.250145149520318$ at $\text{rel_err} = 0.0$ at scale 10^{-8} , and the third branch converges to the pure-electrical Helmholtz eigenvalue $1/a = 5\Xi_{33}/(2H^2\Xi_{11}) = 20497.61$ of (B) (with $\bar{\Xi}_{33} \rightarrow \Xi_{33}$, because e_{33} is scaled to zero together with e_{31}).
- **G_worker**: the worker-process point evaluation reproduces Y_{sense} bit-identically (real and imaginary parts) both at $\omega = 200$ ($Y_{\text{sense}} = 5.00405914569838 \times 10^{-12}$) and at $\omega = 473$, just above the pole ($Y_{\text{sense}} = 6.062573639651388 \times 10^{-10}$).

Methodological note. An earlier pass at G3, scaling only the raw constructor keyword **e31** toward zero while leaving **e33** fixed, appeared to show Q_{in} plateauing near 1.04×10^{-11} instead of vanishing, and the two mechanical branches plateauing near $\lambda \approx \pm 9.557$ instead of the correct $\pm 10.250145 \dots$. Root cause: $\bar{e}_{31} = e_{31} - (c_{13}^E/c_{33}^E)e_{33}$ is dominated by the unscaled **e33** cross term at this material point, so with **e33** held fixed, $\bar{e}_{31}$ plateaus near -8.951 (its value at raw $e_{31} = 0$) and never actually approaches zero – the singular limit being probed was never reached. This was root-caused by direct numerical probe, consistent with this series’ own methodology: precision was ruled out first (**dps** 50/100/150 gave bit-identical wrong answers under the buggy scaling, a real mathematical limit of the wrong quantity, not a conditioning artifact), which is what forced tracing the discrepancy to the test harness rather than the solver. A permanent regression test pins the correct behavior down going forward.

Reported, not gated: sweeping $Y_{\text{sense}}(\omega)$ across the F–F short-circuit fundamental shows the resonance of main-text Figure 3 – magnitude climbing from $\sim 10^{-10}$ well off-resonance to $\pm 9.76 \times 10^{-8}$ within 0.01 rad s^{-1} of 471.3646 , with a clean sign flip through the pole, textbook driven-response behavior for a transfer function with a simple pole at the mechanical resonance and no damping in the operator.

S.8 Cluster job provenance

All jobs on The Mill [4], solver snapshot 2026-07-10.s10. The monolithic piezoelectric path is a new module, unused by the elastic or layered-piezoelectric defaults.

- 2520476 — F–F finite-element queue (**e0/scboth/ oc**), 3/3 RUN COMPLETED, 0 errors, 0 warnings. Mode 2 bit-identical OC/SC at 74.933783 Hz .
- 2520482 — F–F/C–C e_{31} inverse, SENTINEL PASS_ALL, both legs, $\hat{e}_{31} \text{ rel_err } 2.499 \times 10^{-10}$.
- 2521175 — $n = 0$ overtones (F–F/C–C), SENTINEL PASS_ALL, bit-identical to the in-sandbox smoke-test digits.
- 2521176 — thickness sweep (F–F, 9 points), SENTINEL PASS_ALL, **G_found**, **G_theorem**, and **G_longwave** all pass.

- 2521220 — C–F/F–C mixed-edge, SENTINEL PASS_ALL, **G_reduce** bit-identical F–F/C–C reduction, worker-path green.
- 2521701 — force-driven sensing admittance, SENTINEL PASS_ALL, all 5 gates, bit-identical to the in-sandbox run.

Consistent-projection rerun (2026-09-23). Jobs 2520482, 2521175, 2521176, 2521220 and 2521701 used the Gauss-law closure of §S.1. All five probes were re-run unchanged against the corrected solver (default `projection='consistent'`; the old closure is kept as `projection='duan'` and reproduces the old numbers bit-for-bit). Every gate passes (SENTINEL PASS_ALL on all five), and every number in this document and the main text is from that rerun. The rerun was done in-sandbox at the probes' production precision; it was not repeated on the cluster, because every earlier cluster archival run of these same probes reproduced its in-sandbox run bit-for-bit. The finite-element job 2520476 is unchanged: it does not depend on the analytic model.

References

- [1] W.H. Duan, S.T. Quek, Q. Wang, “Free vibration analysis of piezoelectric coupled thin and thick annular plate,” *J. Sound Vib.* 281 (2005) 119–139.
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